



CampusFlow — Database Architecture

Complete Single-Page Database Reference & Entity-Relationship Documentation

Engine: MySQL 8.0+ / MariaDB

Database: campusflow_db

Tables: 31 Total

Charset: utf8mb4

31

TOTAL TABLES

6

USER ROLES

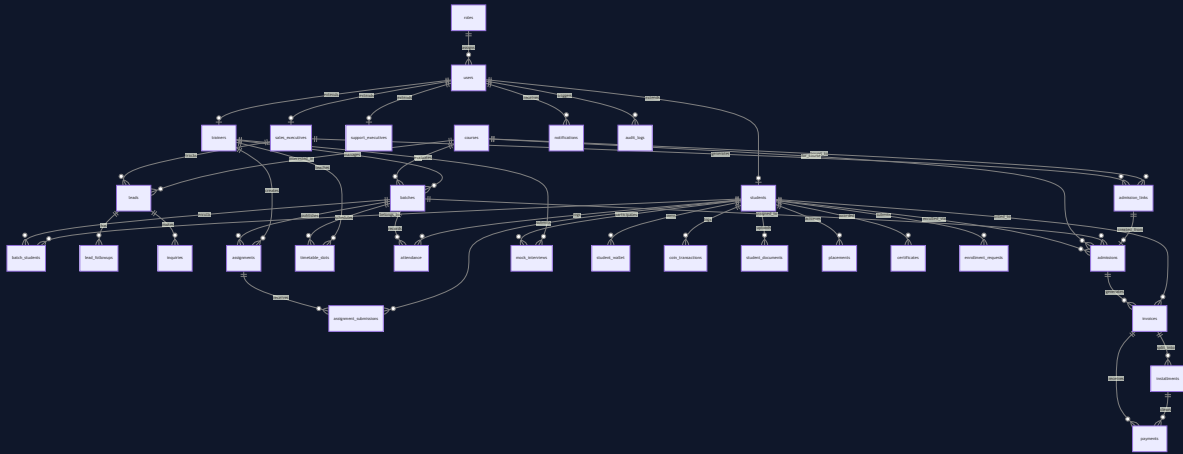
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ACID INNODB COMPLIANT

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FUNCTIONAL MODULES

1. Entity-Relationship (ER) Diagram



2. Complete Database DDL Schema (SQL Script)

```
-- CampusFlow Database Schema
-- Database Name: campusflow_db
-- MySQL Version: 8.0+

CREATE DATABASE IF NOT EXISTS campusflow_db;
USE campusflow_db;

SET FOREIGN_KEY_CHECKS = 0;

DROP TABLE IF EXISTS audit_logs;
DROP TABLE IF EXISTS coin_transactions;
DROP TABLE IF EXISTS student_wallet;
DROP TABLE IF EXISTS enrollment_requests;
DROP TABLE IF EXISTS certificates;
DROP TABLE IF EXISTS placements;
DROP TABLE IF EXISTS student_documents;
DROP TABLE IF EXISTS timetable_slots;
DROP TABLE IF EXISTS lead_followups;
DROP TABLE IF EXISTS notifications;
DROP TABLE IF EXISTS coupons;
DROP TABLE IF EXISTS payments;
DROP TABLE IF EXISTS installments;
DROP TABLE IF EXISTS invoices;
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DROP TABLE IF EXISTS mock_interviews;
DROP TABLE IF EXISTS assignment_submissions;
DROP TABLE IF EXISTS assignments;
DROP TABLE IF EXISTS attendance;
DROP TABLE IF EXISTS admissions;
DROP TABLE IF EXISTS inquiries;
DROP TABLE IF EXISTS leads;
DROP TABLE IF EXISTS batch_students;
DROP TABLE IF EXISTS batches;
DROP TABLE IF EXISTS courses;
DROP TABLE IF EXISTS support_executives;
DROP TABLE IF EXISTS sales_executives;
DROP TABLE IF EXISTS trainers;
DROP TABLE IF EXISTS students;
DROP TABLE IF EXISTS users;
DROP TABLE IF EXISTS roles;

SET FOREIGN_KEY_CHECKS = 1;

-- 1. ROLES TABLE
CREATE TABLE roles (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL UNIQUE,
    description VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 2. USERS TABLE
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    role_id INT NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    phone VARCHAR(20),
    status ENUM('PENDING', 'ACTIVE', 'INACTIVE', 'SUSPENDED') DEFAULT 'PENDING',
    avatar_url VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (role_id) REFERENCES roles(id) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 3. STUDENTS TABLE
CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL UNIQUE,

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roll_number VARCHAR(50) NOT NULL UNIQUE,
dob DATE,
gender ENUM('MALE', 'FEMALE', 'OTHER'),
address TEXT,
qualification VARCHAR(100),
guardian_name VARCHAR(100),
guardian_phone VARCHAR(20),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 4. TRAINERS TABLE

```

CREATE TABLE trainers (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL UNIQUE,
    employee_id VARCHAR(50) NOT NULL UNIQUE,
    specialization VARCHAR(150),
    qualification VARCHAR(100),
    experience_years INT DEFAULT 0,
    bio TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 5. SALES EXECUTIVES TABLE

```

CREATE TABLE sales_executives (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL UNIQUE,
    employee_id VARCHAR(50) NOT NULL UNIQUE,
    target_conversions INT DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

-- 6. SUPPORT EXECUTIVES TABLE

```

CREATE TABLE support_executives (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL UNIQUE,
    employee_id VARCHAR(50) NOT NULL UNIQUE,
    department VARCHAR(100) DEFAULT 'General Support',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 7. COURSES TABLE
CREATE TABLE courses (
    id INT AUTO_INCREMENT PRIMARY KEY,
    code VARCHAR(50) NOT NULL UNIQUE,
    name VARCHAR(150) NOT NULL,
    category VARCHAR(100) DEFAULT 'Web Development',
    description TEXT,
    duration_weeks INT NOT NULL,
    fee_amount DECIMAL(10,2) NOT NULL,
    status ENUM('ACTIVE', 'INACTIVE', 'ARCHIVED') DEFAULT 'ACTIVE',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 8. BATCHES TABLE
CREATE TABLE batches (
    id INT AUTO_INCREMENT PRIMARY KEY,
    course_id INT NOT NULL,
    trainer_id INT,
    batch_code VARCHAR(50) NOT NULL UNIQUE,
    name VARCHAR(150) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    timing VARCHAR(100),
    start_time VARCHAR(20),
    end_time VARCHAR(20),
    room_number VARCHAR(50),
    mode ENUM('ONLINE', 'OFFLINE', 'HYBRID') DEFAULT 'OFFLINE',
    max_students INT DEFAULT 30,
    description TEXT,
    status ENUM('UPCOMING', 'ONGOING', 'COMPLETED', 'CANCELLED', 'INACTIVE') DEFAULT 'UPCOMING',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE RESTRICT,
    FOREIGN KEY (trainer_id) REFERENCES trainers(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 9. BATCH STUDENTS TABLE
CREATE TABLE batch_students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    batch_id INT NOT NULL,
    student_id INT NOT NULL,
    enrolled_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    status ENUM('ENROLLED', 'COMPLETED', 'DROPPED') DEFAULT 'ENROLLED',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,

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        UNIQUE KEY unique_batch_student (batch_id, student_id),
        FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE,
        FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE
    ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 10. LEADS TABLE
CREATE TABLE leads (
    id INT AUTO_INCREMENT PRIMARY KEY,
    sales_exec_id INT,
    candidate_name VARCHAR(100) NOT NULL,
    email VARCHAR(120) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    course_id INT,
    lead_source ENUM('WEBSITE', 'REFERRAL', 'SOCIAL_MEDIA', 'WALK_IN', 'OTHER') DEFAULT 'WEBSITE',
    status ENUM('NEW', 'CONTACTED', 'IN_PROGRESS', 'CONVERTED', 'LOST') DEFAULT 'NEW',
    notes TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (sales_exec_id) REFERENCES sales_executives(id) ON DELETE SET NULL,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 11. INQUIRIES TABLE
CREATE TABLE inquiries (
    id INT AUTO_INCREMENT PRIMARY KEY,
    lead_id INT,
    student_id INT,
    inquiry_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    query TEXT NOT NULL,
    response TEXT,
    status ENUM('PENDING', 'RESOLVED', 'CLOSED') DEFAULT 'PENDING',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (lead_id) REFERENCES leads(id) ON DELETE CASCADE,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 12. ADMISSIONS TABLE
CREATE TABLE admissions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    admission_number VARCHAR(50) NOT NULL UNIQUE,
    lead_id INT,
    student_id INT NOT NULL,
    course_id INT NOT NULL,
    batch_id INT NOT NULL,
    admission_date DATE NOT NULL,
    total_fee DECIMAL(10,2) NOT NULL,

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discount_amount DECIMAL(10,2) DEFAULT 0.00,
final_fee DECIMAL(10,2) NOT NULL,
status ENUM('PENDING', 'CONFIRMED', 'CANCELLED', 'COMPLETED') DEFAULT 'CONFIRMED',
remarks TEXT,
created_by INT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (lead_id) REFERENCES leads(id) ON DELETE SET NULL,
FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE RESTRICT,
FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE RESTRICT,
FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE RESTRICT,
FOREIGN KEY (created_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 13. ATTENDANCE TABLE

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CREATE TABLE attendance (
    id INT AUTO_INCREMENT PRIMARY KEY,
    batch_id INT NOT NULL,
    student_id INT NOT NULL,
    date DATE NOT NULL,
    status ENUM('PRESENT', 'ABSENT', 'LATE', 'LEAVE', 'EXCUSED') DEFAULT 'PRESENT',
    marked_by INT,
    remarks VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    UNIQUE KEY unique_attendance (batch_id, student_id, date),
    FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (marked_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 14. ASSIGNMENTS TABLE

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CREATE TABLE assignments (
    id INT AUTO_INCREMENT PRIMARY KEY,
    course_id INT,
    batch_id INT NOT NULL,
    trainer_id INT NOT NULL,
    title VARCHAR(150) NOT NULL,
    description TEXT,
    instructions TEXT,
    due_date DATETIME NOT NULL,
    deadline DATETIME,
    total_marks INT DEFAULT 100,
    max_marks INT DEFAULT 100,
    status ENUM('DRAFT', 'PUBLISHED', 'CLOSED') DEFAULT 'PUBLISHED',
    file_url VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

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        updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
        FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE CASCADE,
        FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE,
        FOREIGN KEY (trainer_id) REFERENCES trainers(id) ON DELETE CASCADE
    ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 15. ASSIGNMENT SUBMISSIONS TABLE
CREATE TABLE assignment_submissions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    assignment_id INT NOT NULL,
    student_id INT NOT NULL,
    submission_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    submission_text TEXT,
    submission_url VARCHAR(255),
    file_url VARCHAR(255),
    marks_obtained INT,
    feedback TEXT,
    status ENUM('PENDING', 'SUBMITTED', 'LATE', 'REVIEWED', 'GRADED', 'REJECTED') DEFAULT 'PENDING',
    evaluated_by INT,
    reviewed_by INT,
    reviewed_at TIMESTAMP NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    UNIQUE KEY unique_assignment_student (assignment_id, student_id),
    FOREIGN KEY (assignment_id) REFERENCES assignments(id) ON DELETE CASCADE,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (evaluated_by) REFERENCES users(id) ON DELETE SET NULL,
    FOREIGN KEY (reviewed_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 16. MOCK INTERVIEWS TABLE
CREATE TABLE mock_interviews (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    trainer_id INT NOT NULL,
    batch_id INT,
    scheduled_date DATETIME NOT NULL,
    topic VARCHAR(150) NOT NULL,
    score INT,
    status ENUM('SCHEDULED', 'COMPLETED', 'CANCELLED', 'NO_SHOW') DEFAULT 'SCHEDULED',
    feedback TEXT,
    key_strengths TEXT,
    areas_for_improvement TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (trainer_id) REFERENCES trainers(id) ON DELETE CASCADE,

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        FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE SET NULL
    ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 17. INVOICES TABLE
CREATE TABLE invoices (
    id INT AUTO_INCREMENT PRIMARY KEY,
    admission_id INT NOT NULL,
    student_id INT NOT NULL,
    course_id INT,
    invoice_number VARCHAR(50) NOT NULL UNIQUE,
    total_amount DECIMAL(10,2) NOT NULL,
    discount_amount DECIMAL(10,2) DEFAULT 0.00,
    tax_amount DECIMAL(10,2) DEFAULT 0.00,
    net_amount DECIMAL(10,2) NOT NULL,
    paid_amount DECIMAL(10,2) DEFAULT 0.00,
    due_amount DECIMAL(10,2) NOT NULL,
    invoice_date DATE,
    due_date DATE NOT NULL,
    status ENUM('UNPAID', 'PARTIALLY_PAID', 'PAID', 'OVERDUE', 'CANCELLED') DEFAULT 'UNPAID',
    created_by INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (admission_id) REFERENCES admissions(id) ON DELETE CASCADE,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE RESTRICT,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE SET NULL,
    FOREIGN KEY (created_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 18. INSTALLMENTS TABLE
CREATE TABLE installments (
    id INT AUTO_INCREMENT PRIMARY KEY,
    invoice_id INT NOT NULL,
    installment_number INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    paid_amount DECIMAL(10,2) DEFAULT 0.00,
    pending_amount DECIMAL(10,2),
    due_date DATE NOT NULL,
    paid_date DATE,
    payment_mode ENUM('CASH', 'UPI', 'CARD', 'BANK_TRANSFER', 'ONLINE', 'OTHER') DEFAULT 'CASH',
    transaction_id VARCHAR(100),
    status ENUM('PENDING', 'PARTIALLY_PAID', 'PAID', 'OVERDUE') DEFAULT 'PENDING',
    remarks VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (invoice_id) REFERENCES invoices(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 18B. PAYMENTS TABLE
CREATE TABLE payments (
    id INT AUTO_INCREMENT PRIMARY KEY,
    invoice_id INT NOT NULL,
    installment_id INT,
    student_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_date DATE NOT NULL,
    payment_method ENUM('CASH', 'UPI', 'CARD', 'BANK_TRANSFER', 'ONLINE', 'OTHER') DEFAULT 'CASH',
    transaction_reference VARCHAR(100),
    remarks TEXT,
    received_by INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (invoice_id) REFERENCES invoices(id) ON DELETE CASCADE,
    FOREIGN KEY (installment_id) REFERENCES installments(id) ON DELETE SET NULL,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE RESTRICT,
    FOREIGN KEY (received_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 19. COUPONS TABLE
CREATE TABLE coupons (
    id INT AUTO_INCREMENT PRIMARY KEY,
    code VARCHAR(50) NOT NULL UNIQUE,
    discount_type ENUM('PERCENTAGE', 'FIXED') DEFAULT 'PERCENTAGE',
    discount_value DECIMAL(10,2) NOT NULL,
    valid_until DATE,
    max_uses INT DEFAULT 100,
    current_uses INT DEFAULT 0,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 20. NOTIFICATIONS TABLE
CREATE TABLE notifications (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    title VARCHAR(150) NOT NULL,
    message TEXT NOT NULL,
    type ENUM('GENERAL', 'ASSIGNMENT', 'ATTENDANCE', 'INTERVIEW', 'FEE', 'ADMISSION', 'ANNOUNCEMENT') DEFAULT 'GENERAL',
    reference_type VARCHAR(50),
    reference_id INT,
    is_read BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 21. AUDIT LOGS TABLE
CREATE TABLE audit_logs (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    action VARCHAR(100) NOT NULL,
    entity_type VARCHAR(50),
    entity_id INT,
    ip_address VARCHAR(45),
    details TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 22. LEAD FOLLOWUPS TABLE
CREATE TABLE lead_followups (
    id INT AUTO_INCREMENT PRIMARY KEY,
    lead_id INT NOT NULL,
    followed_by INT,
    follow_mode ENUM('CALL', 'WALK_IN', 'EMAIL', 'WHATSAPP', 'OTHER') DEFAULT 'CALL',
    notes TEXT NOT NULL,
    outcome ENUM('INTERESTED', 'NOT_INTERESTED', 'CALLBACK', 'CONVERTED', 'NO_RESPONSE')
    next_followup_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (lead_id) REFERENCES leads(id) ON DELETE CASCADE,
    FOREIGN KEY (followed_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 23. TIMETABLE SLOTS TABLE
CREATE TABLE timetable_slots (
    id INT AUTO_INCREMENT PRIMARY KEY,
    batch_id INT NOT NULL,
    trainer_id INT,
    day_of_week ENUM('MONDAY', 'TUESDAY', 'WEDNESDAY', 'THURSDAY', 'FRIDAY', 'SATURDAY', 'SUNDAY'),
    subject VARCHAR(150) NOT NULL,
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,
    room_number VARCHAR(50),
    notes VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE,
    FOREIGN KEY (trainer_id) REFERENCES trainers(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 24. STUDENT DOCUMENTS TABLE
CREATE TABLE student_documents (

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    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    document_type ENUM('AADHAAR','PAN_CARD','PASSPORT','MARKSHEET_10TH','MARKSHEET_12TH',
    document_name VARCHAR(150) NOT NULL,
    file_path VARCHAR(255),
    status ENUM('PENDING','VERIFIED','REJECTED') DEFAULT 'PENDING',
    remarks VARCHAR(255),
    verified_by INT,
    verified_at TIMESTAMP NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (verified_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 25. PLACEMENTS TABLE

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CREATE TABLE placements (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL UNIQUE,
    batch_id INT,
    course_id INT,
    company_name VARCHAR(150) NOT NULL,
    job_role VARCHAR(150) NOT NULL,
    ctc_lpa DECIMAL(5,2),
    offer_date DATE,
    joining_date DATE,
    placement_type ENUM('FULL_TIME','PART_TIME','INTERNSHIP','FREELANCE') DEFAULT 'FULL_T
    status ENUM('PLACED','SEEKING','NOT_APPLICABLE') DEFAULT 'PLACED',
    offer_letter_url VARCHAR(255),
    notes TEXT,
    recorded_by INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE SET NULL,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE SET NULL,
    FOREIGN KEY (recorded_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

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-- 26. CERTIFICATES TABLE

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CREATE TABLE certificates (
    id INT AUTO_INCREMENT PRIMARY KEY,
    certificate_number VARCHAR(80) NOT NULL UNIQUE,
    student_id INT NOT NULL,
    batch_id INT NOT NULL,
    course_id INT NOT NULL,
    issued_date DATE NOT NULL,

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    valid_until DATE,
    grade VARCHAR(20) DEFAULT 'Pass',
    status ENUM('ISSUED','REVOKED') DEFAULT 'ISSUED',
    issued_by INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE CASCADE,
    FOREIGN KEY (issued_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 27. STUDENT WALLET TABLE
-- Each student has a coin balance. New students receive 10,000 bonus coins on registration
CREATE TABLE student_wallet (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL UNIQUE,
    coins_balance INT NOT NULL DEFAULT 0,
    total_earned INT NOT NULL DEFAULT 0,
    total_spent INT NOT NULL DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 28. COIN TRANSACTIONS TABLE
-- Full audit trail of every coin credit/debit for each student
CREATE TABLE coin_transactions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    type ENUM('CREDIT','DEBIT') NOT NULL,
    coins INT NOT NULL,
    balance_after INT NOT NULL,
    reason VARCHAR(255) NOT NULL,
    reference_type VARCHAR(50),
    reference_id INT,
    created_by INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
    FOREIGN KEY (created_by) REFERENCES users(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 29. ENROLLMENT REQUESTS TABLE
-- Students request to enroll in a course; admin approves/rejects and assigns a batch
CREATE TABLE enrollment_requests (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,

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course_id INT NOT NULL,  
batch_id INT,  
message TEXT,  
status ENUM('PENDING','APPROVED','REJECTED') DEFAULT 'PENDING',  
admin_remarks TEXT,  
coins_deducted INT DEFAULT 0,  
reviewed_by INT,  
reviewed_at TIMESTAMP NULL,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,  
FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE CASCADE,  
FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE SET NULL,  
FOREIGN KEY (reviewed_by) REFERENCES users(id) ON DELETE SET NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```